

Test 1-Section 1



Baldivis
Secondary College

No resources allowed

Name: _____

Class: _____

Time: 20 minutes

Total Mark(1): ____/20

Total Mark(1+2): ____/67

1. Evaluate the following:

[5 marks]

a) 5^2 25 ✓

b) 7^1 7 ✓

c) 12^0 1 ✓

d) 3^{-2} $\frac{1}{9}$ ✓

e) $\frac{1}{2}^{-5}$ 32 ✓

2. Match the following expressions

[5 marks]

$2(6x - 3)$	$12x - 3$ ✓
$-4(1 - 3x)$	$12x - 4$ ✓
$3(3x - 1) + 3x$	$12x - 6$ ✓
$14x - 2(x - 3)$	$12x - 1$ ✓
$2(4 + 3x) + 3(2x - 3)$	$12x + 6$ ✓

3. Circle the expressions which are equivalent to $6a^2b$

[4 marks]

$2a + 3ab$	$\frac{6a^4b^2}{a^2b}$	$8a^2b^2 - 2b$	$2ab \times 3a$
$a^2 \times b + 6b \times a^2$	$2a \times 3ab$	$10a^2b - 4a^2b$	$\frac{18a^2b}{3a^2b}$

4. Expand and simplify the following expressions

[6 marks]

a) $5(2y + 1)$

$$10y + 5 \quad \checkmark$$

b) $7(2 + a) - 4a$

$$= 14 + 7a - 4a$$

$$= 14 + 3a \quad \checkmark$$

c) $p(p + 4)$

$$p^2 + 4p \quad \checkmark$$

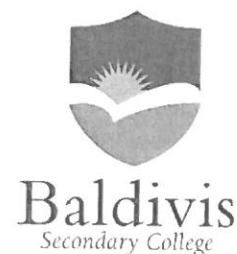
d) $-2b(5b - 3)$

$$-10b^2 + 6b$$

$$\checkmark$$

Year 9

Test 1-Section 2



Resources allowed – Calculator and 1A4 page of notes

Name: _____

Class: _____

Time: 40 minutes

Total Mark: _____/47 46.

Questions 1 to 10 are multiple choice. Circle your answer.

[10 marks]

1. $2^{12} \div 2^7$ is equal to: 2^5
- A 4 B 8 C 16
D 25 E 32
2. $2x^7 \times 7x^3$ is equivalent to:
- A $14x^{21}$ B $14x^{10}$ C $14x^4$
D $9x^{10}$ E $9x^{21}$
3. $-2x^0 + (9x)^0$ simplifies to:
- A -2 B -1 C 0
D 2 E 7
4. $\frac{6a^7b^5}{3a^2b^8}$ simplifies to:
- A $\frac{2a^5}{b^3}$ B $\frac{b^5}{2a^3}$ C $\frac{2a^3}{b^5}$
D $\frac{2b^3}{a^5}$ E $\frac{a^5}{2b^3}$
5. When we expand brackets, we use the distributive law. Which of the following does not represent the distributive law?
- A $a(b + c) = ab + ac$ B $-a(b + c) = -ab - ac$ C $-a(b - c) = -ab - ac$
D $-a(b - c) = -ab + ac$ E $a(b - c) = ab - ac$

6. Which of the following expressions is equivalent to $2(5m + 1)$?

A $7m + 1$

B $10m + 1$

C $10m + 2$

D $12m$

E $11m$

7. Which of the following expressions is equivalent to $5c + 10$?

A $2c \times 3c + 10$

B $5c + c + 10$

C $15c$

D $c + 16 + 4c - 6$

E $2c + 16 - 3c - 6$

8. The expression $4x + 6 + 7x + 12$ can also be written as

A $10x + 19$

B $11x + 18$

C $18x + 11$

D $29x^2$

E $29x$

9. Which expression is equivalent to $3x^2 - 12x + x^3$?

A $3(x^2 - 4x + x^2)$

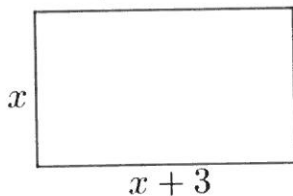
B $3x(x - 4 + x^2)$

C $x^2(3 - 12 + x)$

D $x(3x - 12 + x^2)$

E $3x^2(1 - 4x + x)$

10. Which expression below gives the perimeter of the following shape?



A $x(x + 3)$

B $x + (x + 3)$

C $2(2x + 3)$

D $4x + 12$

E $x^2 + 3x$

11. George invests \$3 500 in a bank that offers an interest rate of 4.5%. He plans to leave the money invested for 2 years.

[3 marks]

a) How much simple interest does George earn?

$3500 \times 0.045 \times 2 = \315

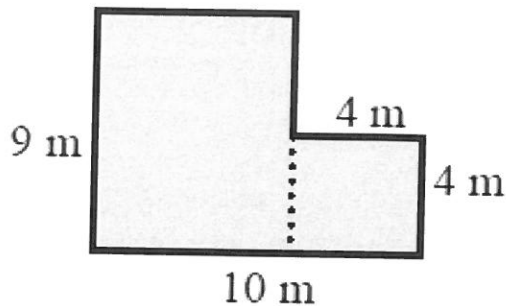
b) What is the total of George's investment after 2 years?

$\$3815$

15. Calculate the area of the following shapes

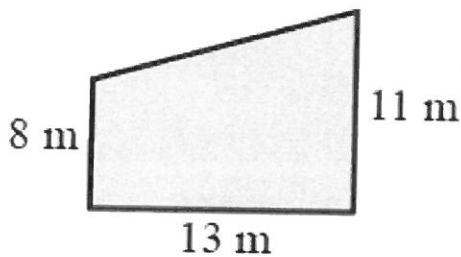
[4 marks]

a)



$$\begin{aligned} 6 \times 9 &= 54 \\ 4 \times 4 &= 16 \\ \text{Total} &= 70 \text{ m}^2 \end{aligned}$$

b)



$$\begin{aligned} \frac{1}{2}(8+11) \times 13 &= 123.5 \text{ m}^2 \\ (-\frac{1}{2} \text{ units}) \end{aligned}$$

16. Expand and simplify the following

[12 marks]

a) $3k(k+1) - 4(2k-5)$

$$\begin{aligned} &= 3k^2 + 3k - 8k + 20 \\ &= 3k^2 - 5k + 20 \end{aligned}$$

b) $(x+3)(x+7)$

$$\begin{aligned} &= x^2 + 3x + 7x + 21 \\ &= x^2 + 10x + 21 \end{aligned}$$

c) $(y-6)(y-5)$

$$\begin{aligned} &= y^2 - 6y - 5y + 30 \\ &= y^2 - 11y + 30 \end{aligned}$$

d) $(p+9)^2$

$$\begin{aligned} &= p^2 + 9p + 9p + 81 \\ &= p^2 + 18p + 81 \end{aligned}$$

12. Simplify each of the following expressions (leave your answers in index form)

4
[5 marks]

a) $12^5 \times 12^3$

12^8 ✓

b) $x^7 \times x^9$

x^{16} ✓

c) $p^8 \div p^5$

p^3 ✓

d) $(5^3)^4$

~~5^7~~
 5^{12} ✓

e) $(y^7)^2$

y^{14} ✓

13. Simplify the following expressions:

[4 marks]

a) $9t^5 \div 3t^3$

$= 3t^2$ ✓

b) $(2k^4)^3$

$= 8k^{12}$ ✓

c) $\frac{30x^5y^6z}{24x^3y^3}$

$= \frac{5x^2y^3z}{4}$ ✓✓

14. Fully factorise the following expressions

[4 marks]

a) $5t + 10$

$= 5(t+2)$ ✓

b) $h^2 - 6h$

$= h(h-6)$ ✓

c) $-2e^2 - 4e$

$= -2e(e+2)$ ✓

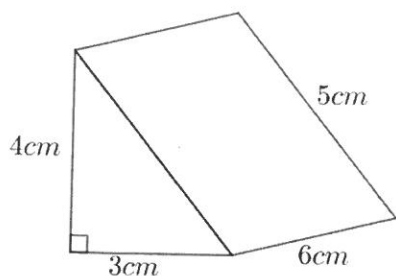
d) $xy^2 + 3x^2y$

$= xy(y+3x)$ ✓

(- 1/2 mark if factorised but not fully)

17. Calculate the Total surface area of the following shape

[3 marks]



$$= 3 \times 4 + (3+4+5) \times 6 \quad \checkmark \checkmark$$

$$= 12 + 72$$

$$= 84 \text{ cm}^2 \quad \checkmark$$

(-1/2 units)

18. How long will it take for an investment of \$8 000 at an interest rate of 3% to earn \$1 200 in simple interest?

[2 marks]

$$t = \frac{I}{Pr} = \frac{1200}{8000 \times 0.03} = 5 \text{ years} \quad \checkmark$$

End of Paper.

